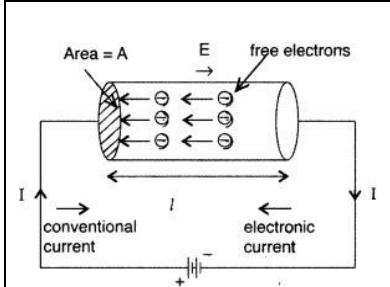
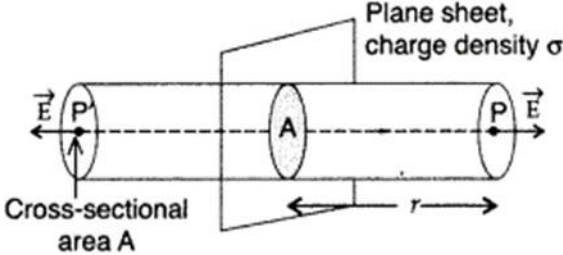
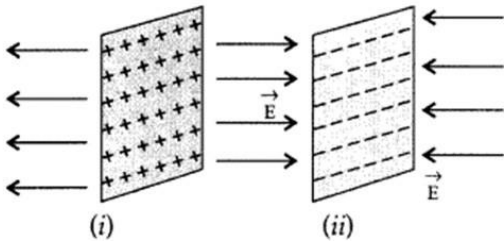
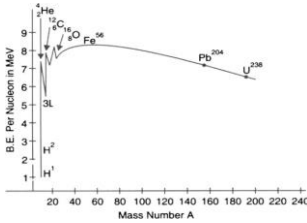


**KENDRIYA VIDYALAYA SANGATHAN**  
**AGRA REGION**  
**PRE – BOARD EXAM 2024-25**  
**CLASS : 12<sup>th</sup> PHYSICS**  
**SET: 3**  
**MARKING SCHEME**

| Q.NO. |                                                                                                                                                                                                                                                                                                                       | MARKS            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1     | (b) infinite                                                                                                                                                                                                                                                                                                          | 1                |
| 2     | b) Charge                                                                                                                                                                                                                                                                                                             | 1                |
| 3     | (a) $E_A > E_B$                                                                                                                                                                                                                                                                                                       | 1                |
| 4     | a) move in a straight-line                                                                                                                                                                                                                                                                                            | 1                |
| 5     | (b) -27.2 eV                                                                                                                                                                                                                                                                                                          | 1                |
| 6     | (a) 1.47                                                                                                                                                                                                                                                                                                              | 1                |
| 7     | (b) Decrease                                                                                                                                                                                                                                                                                                          | 1                |
| 8     | b) 1.227 Å                                                                                                                                                                                                                                                                                                            | 1                |
| 9     | (b) Micro wave                                                                                                                                                                                                                                                                                                        | 1                |
| 10    | (c) 30cm                                                                                                                                                                                                                                                                                                              | 1                |
| 11    | (a) Along the axis of the loop                                                                                                                                                                                                                                                                                        | 1                |
| 12    | (a) 10 V                                                                                                                                                                                                                                                                                                              | 1                |
| 13    | (i) Assertion is correct, reason is correct reason is the correct explanation of Assertion.                                                                                                                                                                                                                           | 1                |
| 14    | (i) Assertion is correct, reason is correct reason is the correct explanation of Assertion.                                                                                                                                                                                                                           | 1                |
| 15    | (iii) Assertion is correct, reason is incorrect                                                                                                                                                                                                                                                                       | 1                |
| 16    | (i) Assertion is correct, reason is correct; reason is a correct explanation for assertion.                                                                                                                                                                                                                           | 1                |
| 17    | It is defined as the number of field lines crossing the surface normally.<br>$\oint \vec{B} \cdot d\vec{S} = B (\Delta S) \cos\theta$ ; It is a scalar; SI unit: Tesla m <sup>2</sup> or Weber                                                                                                                        | 1+1              |
| 18    | For a half-wave rectifier, the output frequency is equal to the input frequency. In this case, the input frequency of the half-wave rectifier is 50 Hz.<br>On the other hand, the output frequency for a full-wave rectifier is twice the input frequency. Therefore, the output frequency is $2 \times 50 = 100$ Hz. | 1<br>1           |
| 19    | PROVE<br><br>OR<br><br>Expression for nuclear density<br><br>1:1                                                                                                                                                                                                                                                      | 2<br><br><br>1+1 |
| 20    | (i) Mobility $\propto 1/\text{Potential difference}$<br>So, if potential difference is doubled, mobility will be halved.<br>(ii) Current density $\propto \text{Potential difference}$<br>So, if potential difference is doubled, current density will also be double.                                                | 2                |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 21 | <p>refractive index of lens 1.25 is less than refractive index of water 1.33. so the nature of lens will change, the lens will behave like a concave lens .</p> <p>OR</p> <p>Power of the lens = +5D</p> <p>Focal length of the lens = <math>f = \frac{1}{D}</math> , so <math>f = \frac{1}{5} \therefore f = 20 \text{ cm}</math></p> <p>Using lens makers' formula, <math>\frac{1}{f_a} = \left( \frac{\mu_g - \mu_a}{\mu_a} \right) \left( \frac{1}{f_1} - \frac{1}{f_2} \right)</math></p> <p><math>\Rightarrow \left( \frac{1}{f_1} - \frac{1}{f_2} \right) = \frac{1}{10} \dots\dots\dots(1)</math></p> <p>, <math>\frac{1}{f_{lq}} = \left( \frac{\mu_g - \mu_{lq}}{\mu_{lq}} \right) \left( \frac{1}{f_1} - \frac{1}{f_2} \right) \Rightarrow \frac{1}{100} = \left( \frac{1.5 - \mu_{lq}}{\mu_{lq}} \right) \left( \frac{1}{10} \right)</math></p> <p><math>\Rightarrow \mu_{lq} = \frac{5}{3}</math></p> | 2                       |
| 22 | <p>(i) (1,2) and (3,4)</p> <p>(ii) (1,3) and (2,4)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>1.5</p> <p>1.5</p>   |
| 23 | <p>(c) Radio wave &lt; (d) visible light &lt; (a) X ray &lt; (b) Gamma ray</p> <p>Greatest penetration power - Gamma ray</p> <p>Greatest ionization power - Radio wave</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>2</p> <p>0.5+0.5</p> |
| 24 | <p>Ans. <math>I_{\max} / I_{\min} = 25/9</math></p> <p><math>I_{\max} / I_{\min} = (a_1 + a_2)^2 / (a_1 - a_2)^2 = 25/9 \Rightarrow a_1 / a_2 = 1/4</math></p> <p><math>I_1 / I_2 = a_1^2 / a_2^2 = (1/4)^2 = 1/16</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>1</p> <p>2</p>       |
| 25 | Statement + Derivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1+2                     |
| 26 | <p>Definition: Drift velocity is defined as the velocity with which free electrons in a conductor get drifted in a direction opposite to the direction of the applied field. Its unit is m/s.</p>  <p>Expression: The magnitude of electric field set up across the conductor is given by</p> <p><math>E = V/l</math></p> <p>Let n be the number of free electrons per unit volume of the conductor. Then, total number of free electrons in the conductor = <math>n \times \text{Volume of the conductor}</math></p>                                                                                                                                                                                                                                                                                                           | <p>1</p> <p>2</p>       |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
|    | <p>Hence, <math>Q = (nAl)e</math></p> <p>Time taken by the charge to cross the conductor length <math>l</math> is given by</p> $t = \frac{1}{v_d} \quad \text{where } [v_d \text{ is drift velocity of electrons}]$ $\therefore I = \frac{Q}{t} = \frac{nAle}{\frac{l}{v_d}} = neAv_d \quad \therefore \boxed{I = neAv_d}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                  |
| 27 | <p>Sol Consider a thin, infinite plane sheet of charge with uniform surface charge density <math>\sigma</math>. We wish to calculate its electric field at a point P at distance <math>r</math> from it</p>  <p>By symmetry, electric field <math>E</math> points outwards normal to the sheet. Also, it must have same magnitude and opposite direction at two points P and F equidistant from the sheet and on opposite sides. We choose cylindrical Gaussian surface of cross-sectional area <math>A</math> and length <math>2r</math> with its axis perpendicular to the sheet. As the lines of force are parallel to the curved surface of the cylinder, the flux through the curved surface is zero. The flux through the plane-end faces of the cylinder is:</p> $\phi_E = EA + EA = 2EA$ <p>Charge enclosed by the Gaussian surface,</p> $q = \sigma A$ <p>According to Gauss's theorem,</p> $\phi_E = \frac{q}{\epsilon_0} \quad \therefore 2EA = \frac{\sigma A}{\epsilon_0} \quad \text{or } E = \frac{\sigma}{2\epsilon_0}$ <p>Clearly, <math>E</math> is independent of <math>r</math>, the distance from the plane sheet.</p>  | 1<br>1<br>1                      |
| 28 | <p>(i) The RMS of the current is 10 A , Current amplitude = 14.14 A</p> <p>(ii) the average power transferred to the circuit in one complete cycle at resonance = 2000 W</p> <p>(iii) At resonance the potential drop across capacitor is zero.</p> <p>OR</p> $\frac{N_S}{N_P} = \frac{E_S}{E_P} = \frac{I_P}{I_S}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0.5+0.5<br>1<br>1<br><br>0.5+0.5 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|    | $\frac{2000}{1000} = \frac{E_S}{120}, E_S = 240 \text{ Volt}$ $\frac{240}{120} = \frac{5}{I_S}, I_S = \frac{5}{2} = 2.5 \text{ Amp.}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>1</p> <p>1</p>                            |
| 29 | Answers: (i) (a) (ii) (c) (iii) (d) (iv) (c) OR (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1+1+1+1                                      |
| 30 | <p>Answer:1 C) 80%</p> <p>Answer: 2 B) 100 Hz</p> <p>Answer3 : C) To provide isolation</p> <p>Answer4: D) All of the above</p> <p>OR</p> <p>Answer4 : D) To provide a path for current flow</p>                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> |
| 31 | <p>Levelled dig.+Prove</p> <p>OR</p> <p>derivation +Definition+ direction of force</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>2+3</p> <p>3+1<br/>+0.5+0.5</p>           |
| 32 | <p>The total energy required to disintegrate the nucleus into its constituent particles is called binding energy of the nucleus.</p>  <p>Salient features of the curve</p> <p>(i) The intermediate nuclei have a large value of binding energy per nucleon, so they are most stable. (For <math>30 &lt; A &lt; 63</math>)</p> <p>(ii) The binding energy per nucleon has low value for both the light and heavy nuclei. So, they are unstable nuclei.</p> <p>OR</p> <p>Definition of nuclear force.</p> <p>Graph +2 conclusion</p> | <p>2+2+1</p> <p>1+ 2+2</p>                   |
| 33 | <p>Conditions</p> <p>(i)Increase + reasons (ii) decrease + reasons (iii) increase + reasons</p> <p>OR</p> <p>Definition +graph+ derivation for width of central maxima</p>                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>2+3</p> <p>1+2+2</p>                      |